Python Project

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Data Processing Task

1- Loading data using Pandas

```
import pandas as pd
df  = pd.read_excel(r"C:\Users\HP\Downloads\
            ProjectData.xlsx")
print(df)
```

105 Rows -- 8 columns (Total rows and coulms in data)

2- Checking Missing Values

```
import pandas as pd
df.isnull( ).sum( )
```

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10
1	1002	2025-01-02	West	Phone	Electronics	10	500	5
2	1003	2025-01-03	North	Chair	Furniture	8	120	0
3	1004	2025-01-04	South	Table	Furniture	3	300	15
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5
...	...	...	...	...	...	...	...	...
100	1101	2025-04-11	East	Tablet	Electronics	11	335	7
101	1102	2025-04-12	West	Printer	Electronics	6	240	5
102	1103	2025-04-13	North	Bookshelf	Furniture	4	200	10

```
OrderID          0
Date             0
Region           0
Product          0
Category         0
Units_Sold       0
Unit_Price       0
Discount (%)     0
dtype: int64
```

3- Creating new colums

1 Total Sales

```
df["Total_Sales"]=df["Units_Sold"] * df["Unit_Price"]
df.head()
```

2- Discount Amount

```
df["Discount_Amount"] = (df["Total_Sales"] * df["Discount (%)"])/100df.head()
df.head()
```

3- Final Sales

```
df["Final_Sales"]=df["Total_Sales"] - df["Discount_Amount"]
df.head()
```

	OrderID	Date	Region	Product	Category	Units_Sold	Unit_Price	Discount (%)	Total_Sales	Discount_Amount	Final_Sales
0	1001	2025-01-01	East	Laptop	Electronics	5	800	10	4000	400.0	3600.0
1	1002	2025-01-02	West	Phone	Electronics	10	500	5	5000	250.0	4750.0
2	1003	2025-01-03	North	Chair	Furniture	8	120	0	960	0.0	960.0
3	1004	2025-01-04	South	Table	Furniture	3	300	15	900	135.0	765.0
4	1005	2025-01-05	East	Headphones	Electronics	15	100	5	1500	75.0	1425.0

Analysis Task

1- **Total Revenue**

```
Total_Revenue = df["Final_Sales"].sum()
print("Total_Revenue = " , Total_Revenue)
```

2- **Region Which has Highest Sales**

```
Region_Sales = df.groupby("Region")["Final_Sales"].sum( )
print(Region_Sales)
print("Highest_Region = " , Region_Sales.idxmax( ))
```

3- **Category Which Performs Best**

```
Category_Sales=df.groupby("Category")["Final_Sales"].sum( )
print(Category_Sales)
Best_Category = Category_Sales.idxmax()
print("Best Performing Category =" , Best_Category)
```

4 – **Average Discount**

```
Average_Discount =df["Discount (%)"].mean()
print("Average_Discount =" , Average_Discount)
```

```
Total_Revenue =  201513.65000000002
```

```
Region
East      51716.10
North     43602.70
South     52256.55
West      53938.30
```

```
Highest_Region =  West
```

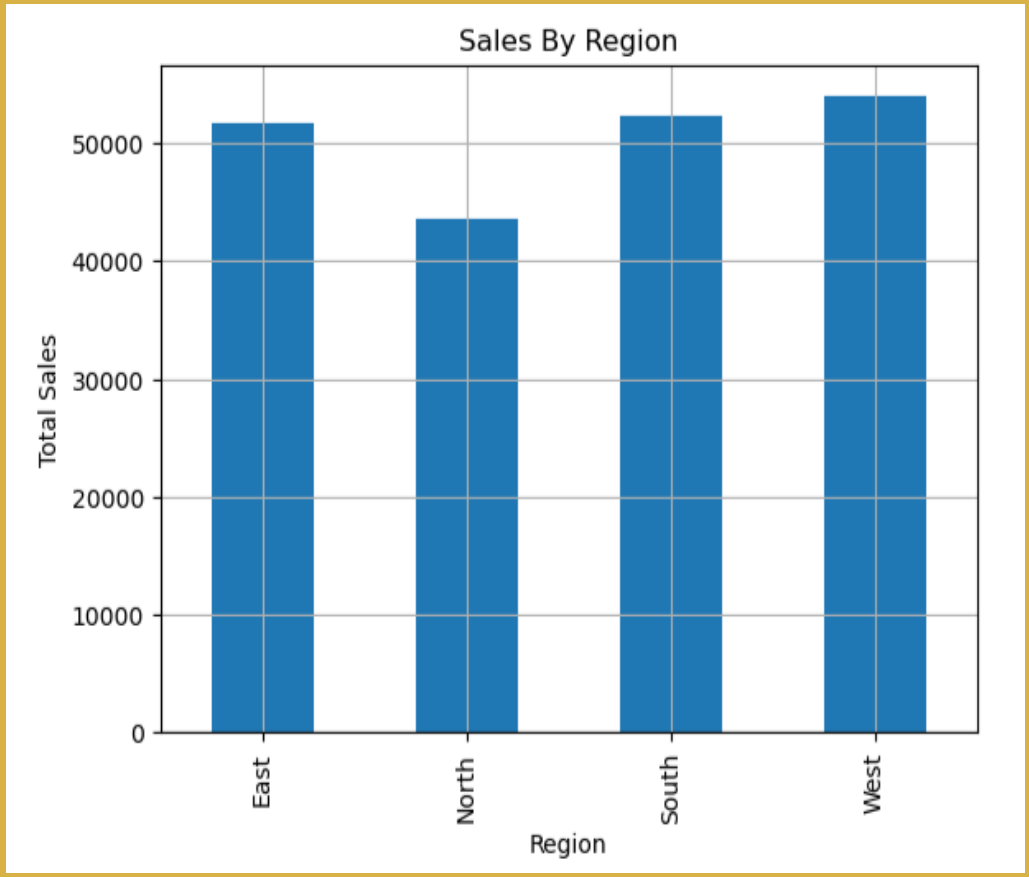
```
Category
Electronics    147162.85
Furniture       54350.80
Name: Final_Sales, dtype: float64
Best Performing Category = Electronics
```

```
Average_Discount = 7.485714285714286
```

Visualization Task

1- **Sales By Region [Bar Chart]**

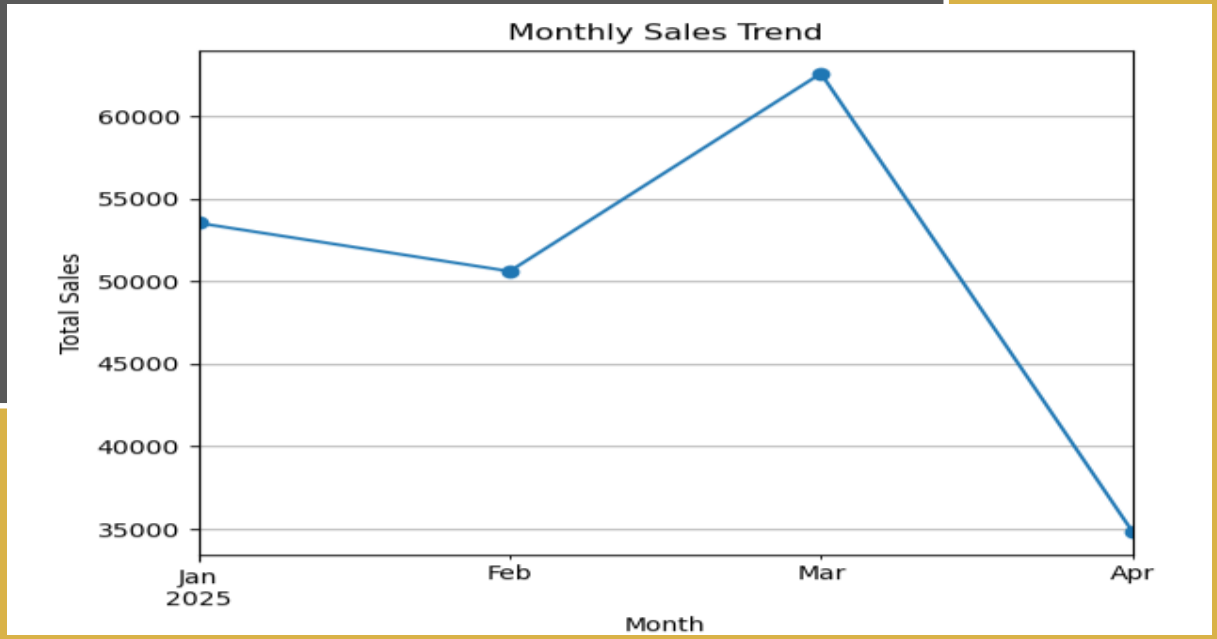
```
import matplotlib.pyplot as plt
Region_Sales=df.groupby("Region")["Final_Sales"].sum()
 Region_Sales.plot(kind="bar")
 plt.title("Sales By Region")
plt.xlabel("Region")
plt.ylabel("Total Sales")
plt.grid(True)
plt.show
```



Visualization Task

2- **Sales Trend Over Time [Line Chart]**

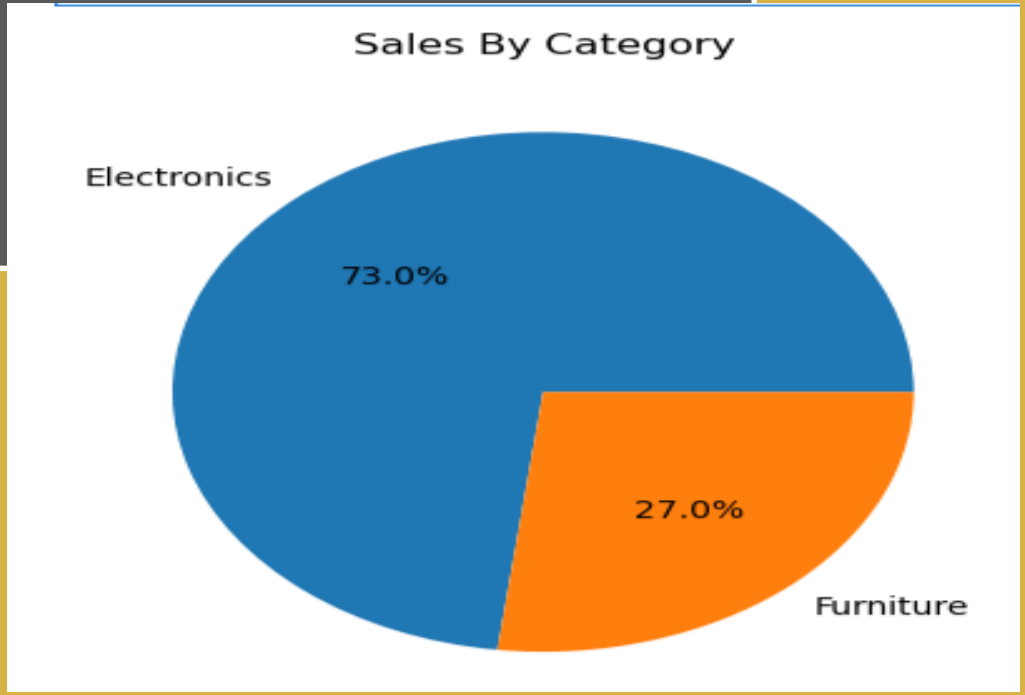
```
Monthly_Sales = df.groupby(df["Date"].dt.to_period("M"))["Final_Sales"].sum( )
Monthly_Sales.index = Monthly_Sales.index.to_timestamp()
Monthly_Sales.plot(marker="o")
plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")
plt.grid(True)
plt.show
```



Visualization Task

3 - Category Distribution [Pie Chart]

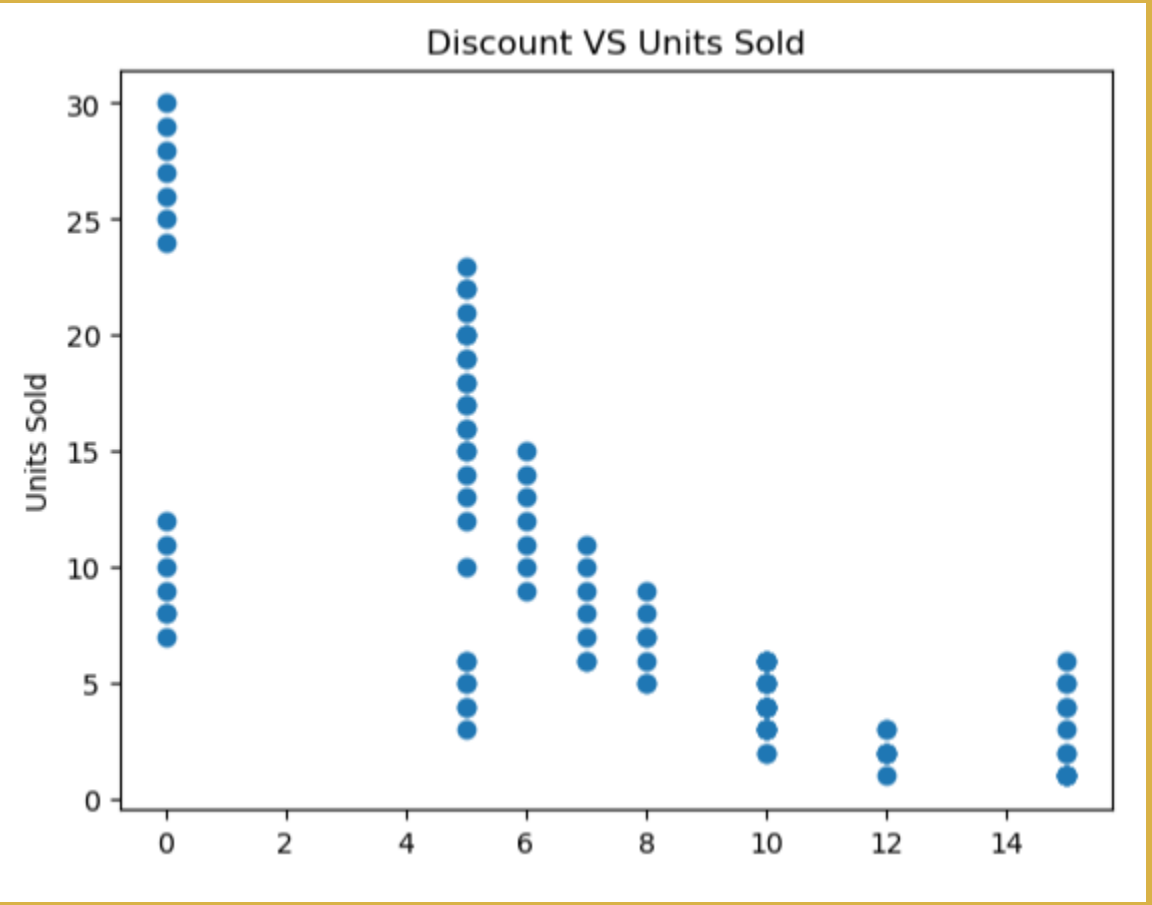
```
Category_Sales = df.groupby("Category")["Final_Sales"].sum()
Category_Sales.plot(kind="pie",autopct="%1.1f%%")
 plt.title("Sales by category")
 plt.ylabel("")
 plt.show
```



Visualization Task

4 – **Discount VS Units Sold [Scatter Plot]**

```
plt.scatter(df["Discount (%)"],df["Units_Sold"]
plt.title("Discount VS Units Sold")
plt.xlabel("Discount(%)")
plt.ylabel("Units Sold")
plt.show
```



Numpy Usage Task

Calculating
[Mean Sales , Standard Deviation , Variance]

```
import numpy as np
Sales = df["Final_Sales"].values
print("Mean =",np.mean(Sales))
print("Variance =",np.var(Sales))
print("Standard Deviation =",np.std(Sales))
```

```
Mean = 1919.1776190476194
Variance = 2431162.4845229024
Standard Deviation = 1559.218549313374
```

Numpy Usage Task

Normalizing Sales Data Using Numpy

```
import numpy as np
Sales = df["Final_Sales"].values
Sales_min = np.min(Sales)
Sales_max = np.max(Sales)
normalized_sales = (Sales - Sales_min) / (Sales_max -Sales_min)
df["Normalized_Sales"] = normalized_sales
print(df[["Final_Sales","Normalized_Sales"]].head())
```

	Final_Sales	Normalized_Sales
0	3600.0	0.407054
1	4750.0	0.556731
2	960.0	0.063450
3	765.0	0.038070
4	1425.0	0.123971

Business Insights Summary

🏆 **Regional Performance**

The **West region generated the highest total sales**, making it the top-performing region.
This indicates strong demand and market presence in this area.
The **South region showed comparatively lower performance**, suggesting potential for improvement through marketing or sales strategies.
👉 Business Action: Focus on strengthening low-performing regions and maintaining growth in high-performing ones.

✅ **Final Business Conclusion**

The business performs strongest in the **West region and Electronics category**.
Sales vary across time and transactions, with some extreme values observed.
Improving weaker regions, optimizing discounts, and focusing on high-demand products can increase overall profitability